School of Future Tech

Case Study Report

on

HealConnect – Online Physiotherapy Platform

by

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1. Introduction to the Case Study

HealConnect is a secure, web-based online physiotherapy platform connecting patients with certified physiotherapists via real-time video consultations. It provides accessible, affordable, high-quality home-based rehabilitation, overcoming traditional clinic barriers.

This case study details the design and front-end development (HTML5, CSS3, vanilla JavaScript) of the multi-page, responsive web application. The platform features five interconnected pages: Home, Booking, Exercise Library, Patient Progress Dashboard, and Pricing, forming a cohesive telehealth experience. It supports recovery from conditions like back and knee injuries, shoulder problems, and post-surgical rehabilitation, highlighting web development application in healthcare with a focus on UX, accessibility, and responsive design.

2. Problem Statement / Case Background (Abstract)

Background

Quality physiotherapy is often inaccessible due to long waits, high travel costs, and limited specialist availability. Traditional models struggle to maintain the consistent, scheduled interactions vital for physical rehabilitation. The COVID-19 pandemic highlighted this fragility, boosting the need for remote rehabilitation.

A growing demand exists for an intuitive, trustworthy platform offering not only video consultations but also structured recovery programs, progress tracking, exercise libraries, and seamless appointment booking.

Abstract

This case study presents the design and implementation of HealConnect, a fully functional online physiotherapy web application. The platform enables patients to discover specialist therapists, book video sessions, access a curated exercise library, track their recovery progress, and manage appointments through a personal dashboard. Built with HTML5, CSS3, and JavaScript, the application demonstrates the integration of modern front-end development techniques including custom CSS variables, responsive grid layouts, JavaScript-driven interactivity, real-time form validation, and sessionStorage-based data persistence. The design prioritises clarity, accessibility, and professional medical aesthetics to build patient trust and encourage consistent engagement throughout the recovery journey.

3. Case Study Design

The HealConnect platform is designed as a modular, multi-page web application with the following structural components:

Information Architecture

- index.html – Landing page with hero section, services overview, featured therapists, testimonials, and a CTA banner.
- book.html – Multi-field appointment booking form with therapist selection cards, time-slot picker, and a live booking summary sidebar.
- library.html – Searchable, filterable exercise library with category-based cards and video modal placeholders.
- progress.html – Patient dashboard with session statistics, upcoming and completed session lists, a recovery progress bar, and a personal notes journal.
- pricing.html – Three-tier pricing page with a per-session/monthly billing toggle, feature comparison table, and CTA.

Visual Design System

- A consistent design language is applied across all pages using a shared styles.css file with CSS custom properties (variables) for colour, spacing, typography, and border radii.
- Primary palette: forest green for brand identity, white for backgrounds, and dark charcoal for body text.
- Typography: Inter font family for a clean, modern medical aesthetic.

Responsive Layout

- CSS Grid and Flexbox are used throughout to ensure proper rendering across desktop, tablet, and mobile breakpoints.
- A hamburger menu with a mobile slide-in overlay is implemented for navigation on small screens.

Interactivity & State Management

- All user interactions — booking form validation, therapist selection, time slot picking, pricing toggle, session completion, and note saving — are handled by a centralised main.js file.
- State is persisted across page interactions using sessionStorage.

4. Methods & Algorithms / Technology Applied in the Problem Statement / Case Study

Key methods and technologies applied:

Front-End Technologies

- HTML5 for semantic document structure and accessible markup (ARIA roles, labels, and landmarks).
- CSS3 with custom properties, keyframe animations (fade-up, shimmer), pseudo-elements, and media queries.
- Vanilla JavaScript (ES6+) for DOM manipulation, event handling, and dynamic UI updates.

Form Validation Algorithm

- Custom client-side validation checks all required fields before submission: name, email (regex), phone, date (no past dates), injury category, therapist selection, and time slot.
- Inline error messages are surfaced next to each invalid field with smooth visual transitions.

Booking Summary Engine

- Real-time event listeners on the booking form update a live summary sidebar with patient name, date, time, therapist, condition, session mode, and dynamic session fee calculation (including home-visit surcharge).

Exercise Library Filtering

- Category-based filter buttons apply CSS class toggling and JavaScript DOM filtering to show or hide exercise cards matching the selected condition category.
- A search input further filters visible cards by title keyword in real time.

Progress Dashboard Logic

- Mark-as-done buttons on upcoming sessions update session status badges, decrement upcoming count, increment completed count, and animate the progress bar.
- Recovery notes are saved to sessionStorage and dynamically rendered as timestamped note cards.

Technology Stack Used:

- Languages: HTML5, CSS3, JavaScript (ES6+).
- Libraries and tools: None — pure vanilla implementation with no external JavaScript frameworks.
- Fonts: Google Fonts (Inter).
- Environment: Any modern browser; development via VS Code with Live Server.
- Data storage: sessionStorage for ephemeral client-side persistence.

5. Problem Statement / Case Study Implementation Details and Snapshots

The implementation is organised into three core files shared across all pages:

styles.css

- Global CSS variables for the design system (colours, spacing, radii, shadows).
- Component styles: navbar, hero, cards, forms, booking layout, pricing toggle, dashboard stats, exercise library grid, footer.
- Keyframe animation definitions: fadeUp, shimmer, pulse.
- Responsive breakpoints at 768px and 480px for tablet and mobile layouts.

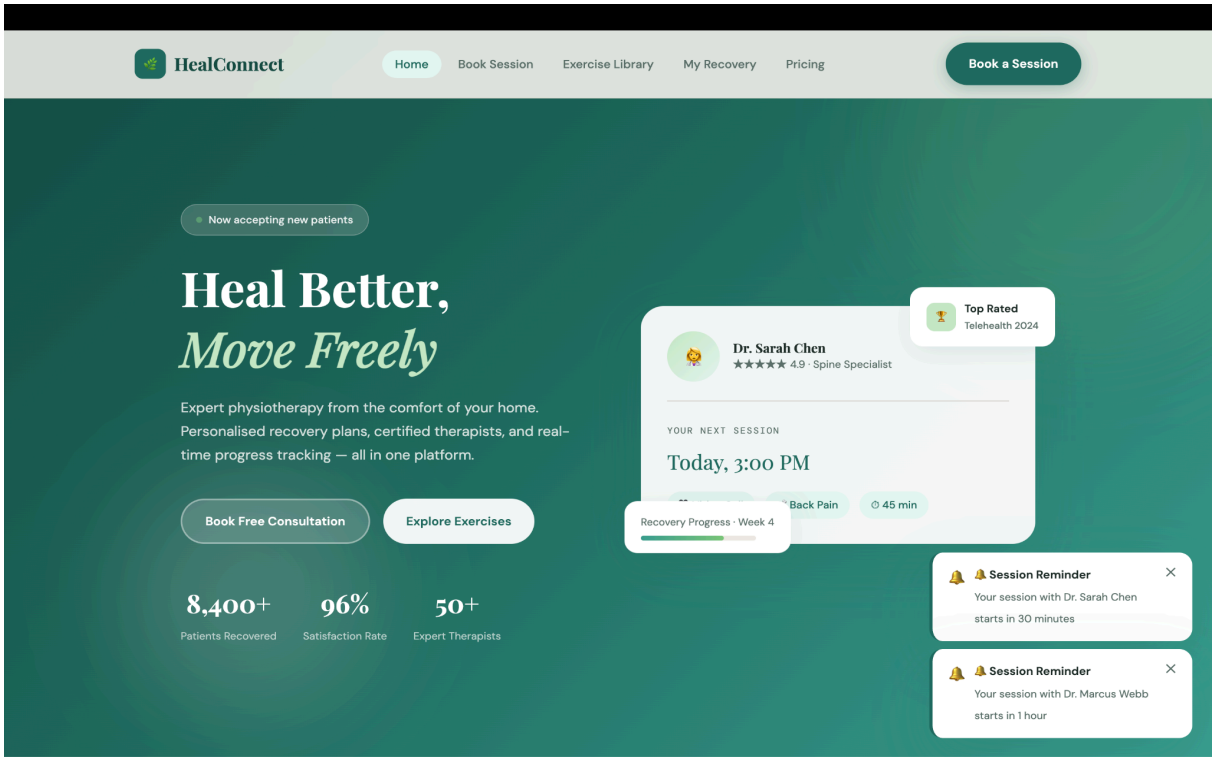
main.js

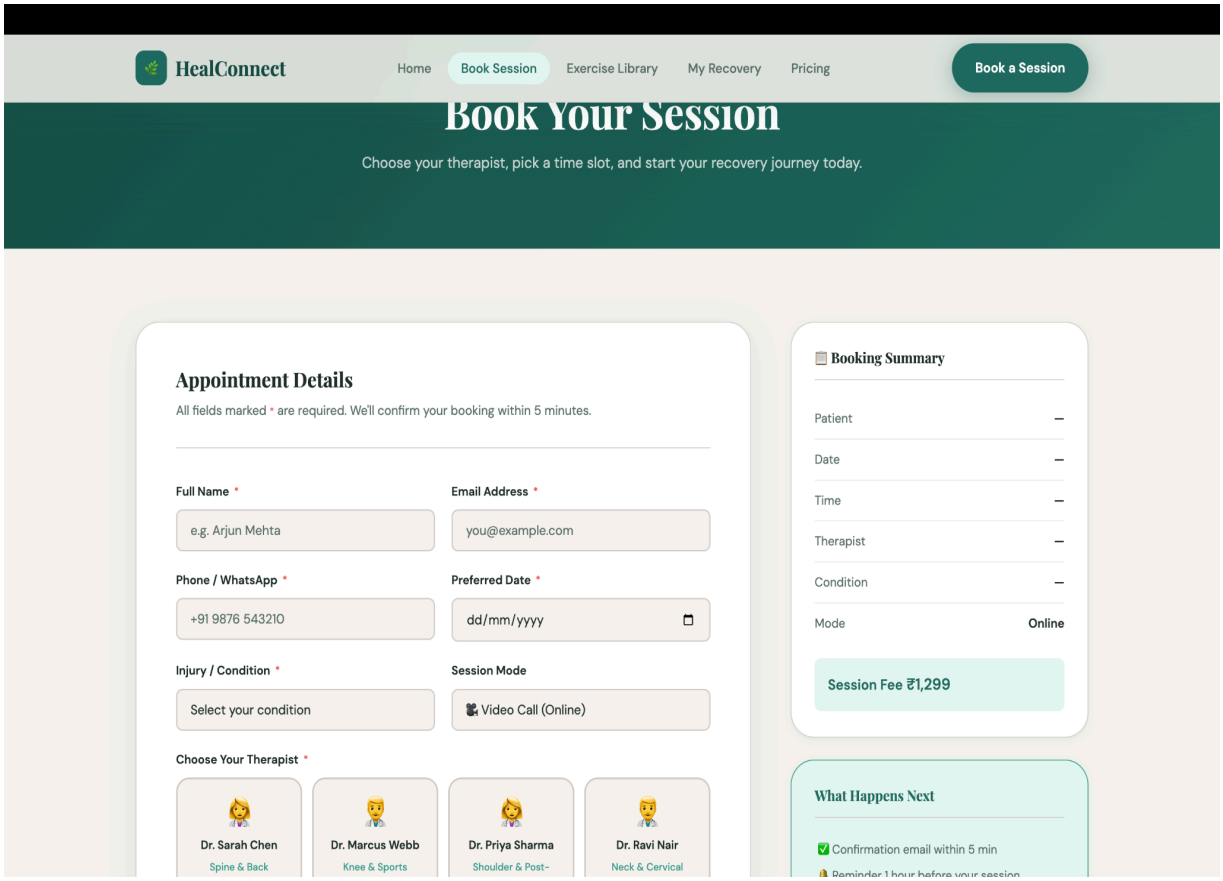
- Mobile menu open/close logic.
- Intersection Observer-based fade-up scroll animation trigger.
- Booking form: therapist card selection, time slot selection, live summary updates, fee calculator, and full validation on submit.
- Pricing page: billing toggle switching between per-session and discounted monthly prices.
- Progress page: mark-done functionality, note saving, and sessionStorage retrieval.
- Notification bar: reusable toast notification system used across all pages.

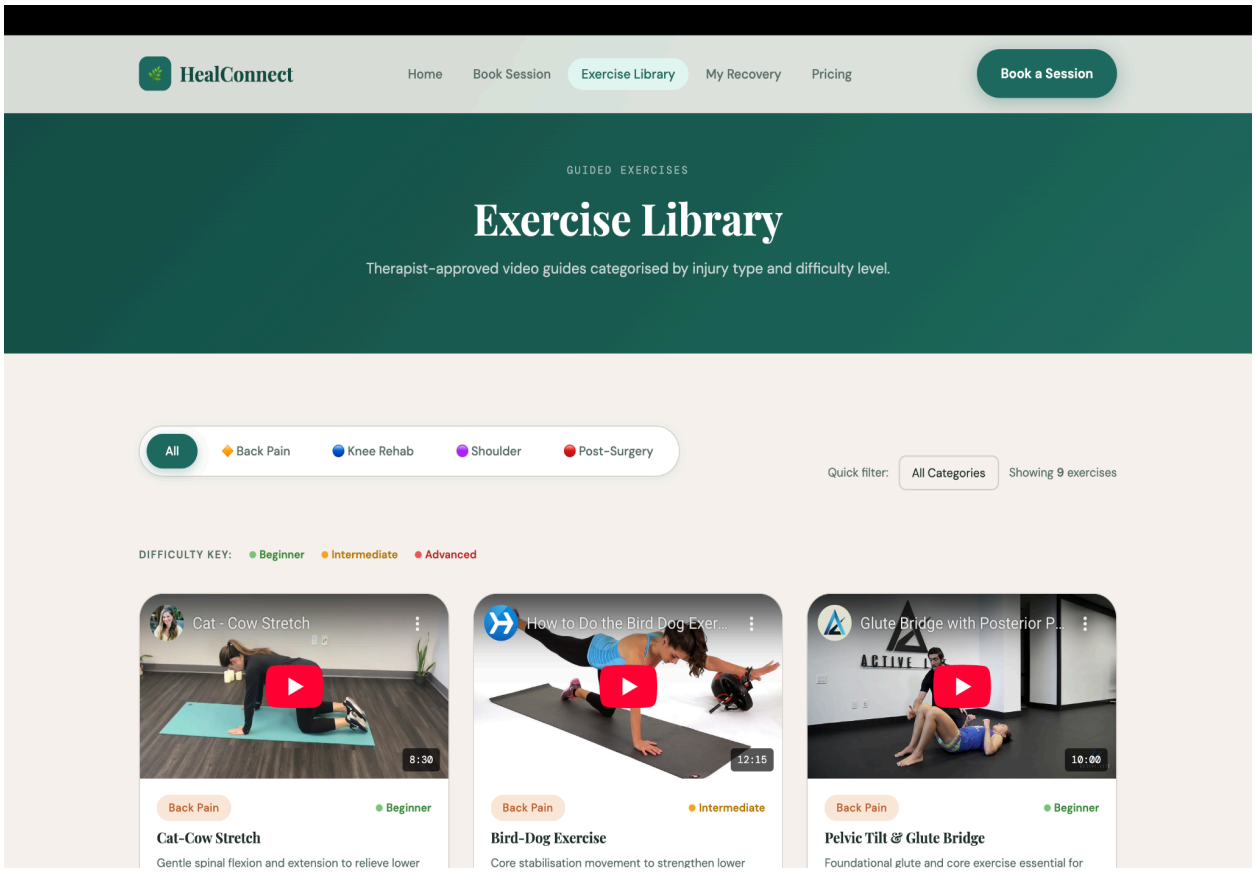
Individual HTML Pages (index, book, library, progress, pricing)

- Each page uses consistent semantic HTML structure: <nav>, <header>, <main>, <section>, <aside>, <footer>.
- ARIA labels, roles, and landmarks are used throughout for screen-reader accessibility.

Snapshots to include:






HealConnect
Home
Book Session
Exercise Library
My Recovery
Pricing
Book a Session
GUIDED EXERCISES
Exercise Library
Therapist-approved video guides categorised by injury type and difficulty level.
All
Back Pain
Knee Rehab
Shoulder
Post-Surgery
Quick filter:
All Categories
Showing 9 exercises
DIFFICULTY KEY:
Beginner
Intermediate
Advanced
Cat - Cow Stretch
8:30
Back Pain
Beginner
Cat-Cow Stretch
Gentle spinal flexion and extension to relieve lower
How to Do the Bird Dog Exer...
12:15
Back Pain
Intermediate
Bird-Dog Exercise
Core stabilisation movement to strengthen lower
Glute Bridge with Posterior P...
10:00
Back Pain
Beginner
Pelvic Tilt & Glute Bridge
Foundational glute and core exercise essential for

HealConnect

Home | Book Session | Exercise Library | My Recovery | Pricing

Book a Session

PATIENT DASHBOARD

My Recovery

Track your sessions, monitor progress, and log your recovery notes.

DASHBOARD

- Overview
- Sessions
- Notes
- Exercises
- Settings

Recovery Week 4

72% programme complete

8	3	72%	4.9
Sessions Completed	Upcoming Sessions	Recovery Progress	Avg. Session Rating

Upcoming Sessions

05 MAR — **Back Pain Therapy**
Dr. Sarah Chen · 3:00 PM · 45 min · Video Call — Upcoming

10 MAR — **Knee Rehabilitation**
Dr. Marcus Webb · 11:00 AM · 60 min · Video Call — Upcoming — Mark Done

Session Reminder
Your session with Dr. Sarah Chen starts in 30 minutes

Session Reminder
Your session with Dr. Marcus Webb starts in 1 hour

HealConnect

Home | Book Session | Exercise Library | My Recovery | Pricing

Book a Session

Per Session — Monthly Plan — Save up to 25%

Select this plan

STARTER
Essential Care
Rs. ₹799
per session

- 1 live video session
- Basic exercise guide (5 videos)
- Session summary notes
- Email support
- Progress tracking
- Custom recovery plan
- Priority booking

Get Started

MOST POPULAR

Select this plan

PLUS
Complete Recovery
Rs. ₹1,299
per session

- 1 live video session
- Full exercise library access
- Detailed progress tracking
- Custom recovery plan
- Session reminders and notes
- Chat support between sessions
- Priority booking

Choose Plus

Select this plan

PREMIUM
Elite Rehabilitation
Rs. ₹1,899
per session

- 1 live video session (60 min)
- Full exercise library access
- Advanced progress analytics
- Custom recovery
- Priority therapist
- Unlimited chat su
- Home visit optio

Go Pr

Session Reminder
Your session with Dr. Sarah Chen starts in 30 minutes

Session Reminder
Your session with Dr. Marcus Webb starts in 1 hour

6. Problem Statement / Case Study Results and Conclusion

Findings

- A fully responsive, five-page web application was successfully built and tested across modern browsers (Chrome, Firefox, Safari, Edge) and screen sizes.
- The booking form with real-time validation and live summary significantly improves the user experience by reducing form submission errors and providing immediate feedback.
- The exercise library filtering system allows patients to quickly locate relevant exercises by condition, reducing cognitive load.
- The progress dashboard's mark-done functionality and sessionStorage-based notes provide a lightweight but effective patient engagement mechanism without requiring a backend database.
- Consistent application of the CSS design system across all five pages results in a cohesive brand experience, contributing to perceived trustworthiness — a critical factor in healthcare platforms.

Conclusion

This case study details the front-end development of HealConnect, an accessible, user-friendly online physiotherapy platform. The project uses semantic HTML, advanced CSS, and JavaScript. It establishes a strong base for future additions like a backend API for appointment management, user authentication, video-call integration (e.g., WebRTC), and a mobile app. HealConnect shows that effective UI/UX and clean code can create a compelling telehealth product addressing patient needs.

7. References

- Mozilla Developer Network (MDN) – Official documentation for HTML5, CSS3, and JavaScript APIs. https://developer.mozilla.org
- W3C Web Accessibility Initiative (WAI) – ARIA roles, landmark guidance, and accessible form patterns. https://www.w3.org/WAI
- Google Fonts – Inter typeface used throughout the application. https://fonts.google.com/specimen/Inter
- CSS-Tricks – Guides on CSS Grid, Flexbox, custom properties, and animation techniques. https://css-tricks.com
- Web.dev by Google – Best practices for responsive design, performance, and progressive web applications. https://web.dev
- HIPAA Journal – Reference for data privacy and security compliance considerations in telehealth platforms. https://www.hipaajournal.com
- Academic articles and tutorials on telehealth UX design and online physiotherapy platform architecture.